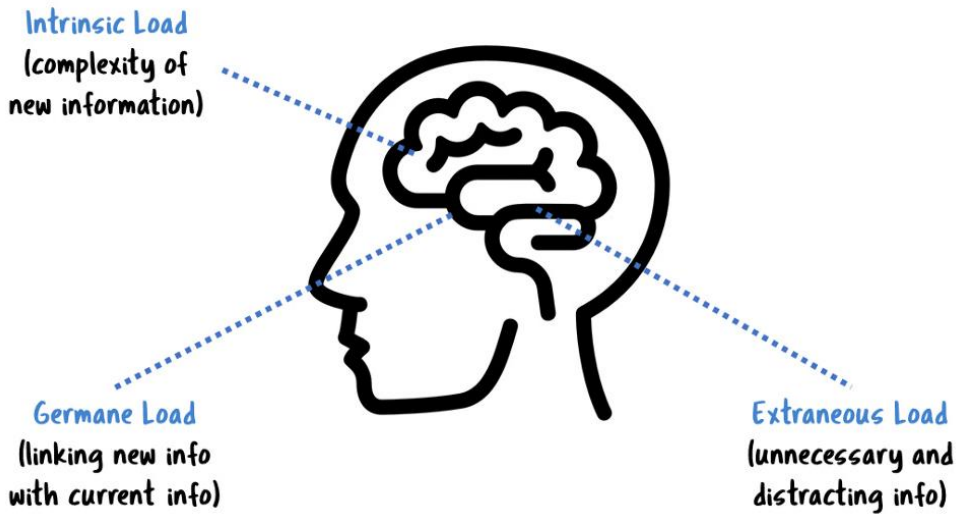
Same Facts, Different Interfaces: Evaluating FAQ, Retrieval Chatbot, and AI Chatbot for Student Services

Henry Kofi Dodzi Dakpo (Presenter)
BA Psychology & Information Studies
University of Ghana, Legon

CoE International Conference 2026

Background

- Universities use FAQ webpages and chatbots for student information.
- Finding relevant answers can demand a lot of mental effort.
- Interfaces may differ in usability, speed, accuracy and trust.





Prior Work and Research Gap

- Han & Lee (2022): FAQ chatbot users reported greater barriers than webpage users.
- Suherwin et al. (2026): RAG outperformed an FAQ chatbot in accuracy but responded more slowly.
- Stadler et al. (2024): ChatGPT reduced effort but weakened scientific justifications.
- Limited three-interface comparisons using shared student-service information.



Goal & Objectives

Goal

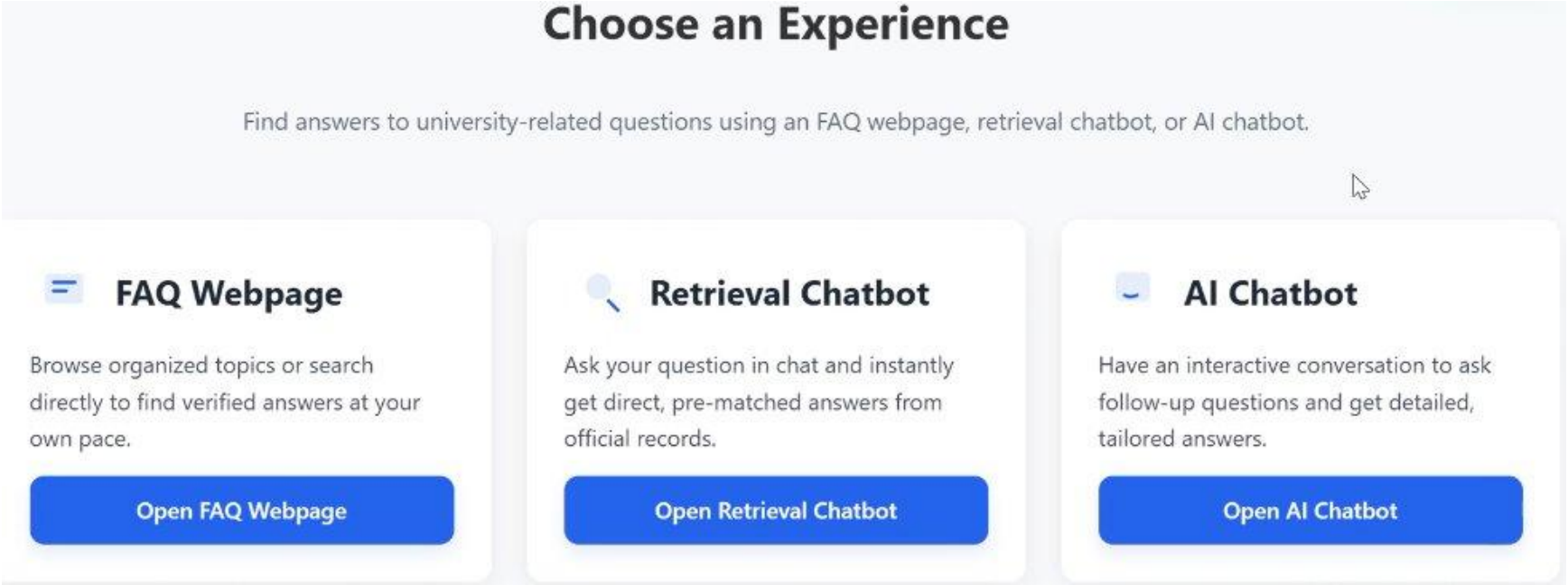
- To compare the user experience of three student-service interfaces using a shared knowledge base.

Objectives

- Develop an FAQ webpage, retrieval chatbot, and AI chatbot using the same student-service knowledge base.
- Compare perceived speed, mental effort, ease of use, clarity, trust, accuracy, and satisfaction across the interfaces.
- Identify users' preferred interface and the reasons for their preferences.

Methodology

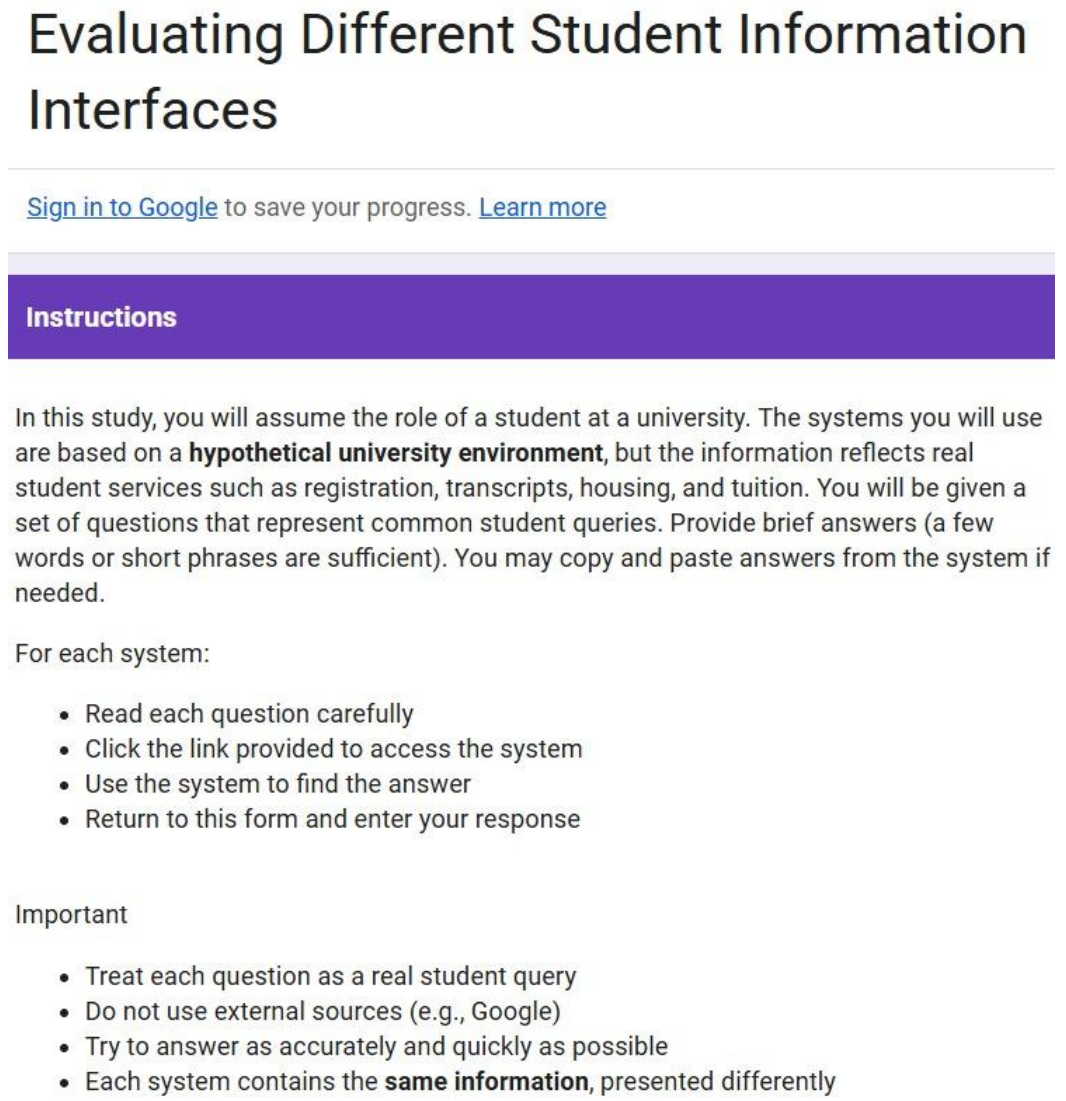
- Developed three systems:
  - FAQ webpage
  - Retrieval-based chatbot
  - AI-powered chatbot
- Shared knowledge base for a hypothetical university.
- Used the shared knowledge-base across all systems
- Iterative Testing and Refinement
- Deployment and Evaluation



Methodology

- **Study Procedure**
  - Participants interacted with all three systems independently.
  - Asked the same questions across systems.
  - Completed a short survey after each system.

- **Measures**
  - Ease of use, mental effort, accuracy, and trust.



Evaluating Different Student Information Interfaces

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Instructions

In this study, you will assume the role of a student at a university. The systems you will use are based on a **hypothetical university environment**, but the information reflects real student services such as registration, transcripts, housing, and tuition. You will be given a set of questions that represent common student queries. Provide brief answers (a few words or short phrases are sufficient). You may copy and paste answers from the system if needed.

For each system:

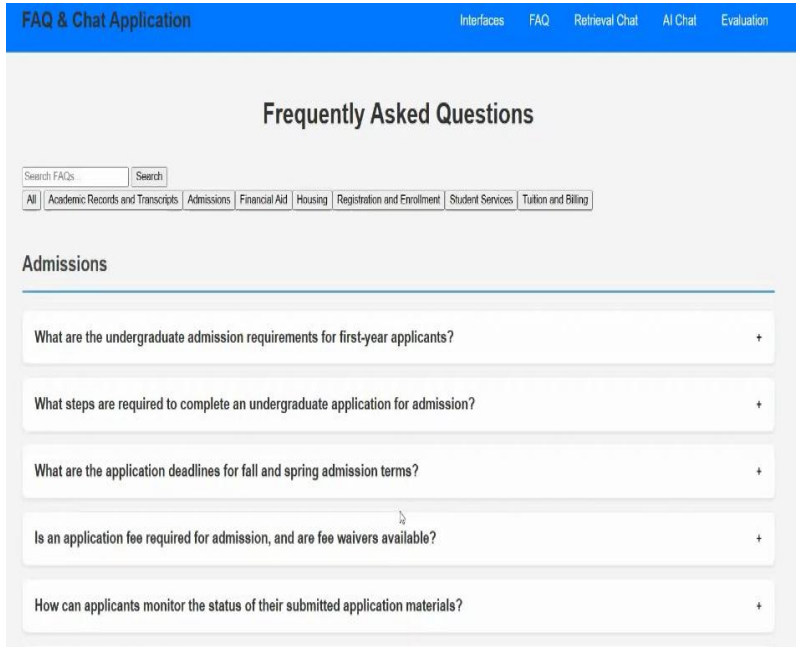
- Read each question carefully
- Click the link provided to access the system
- Use the system to find the answer
- Return to this form and enter your response

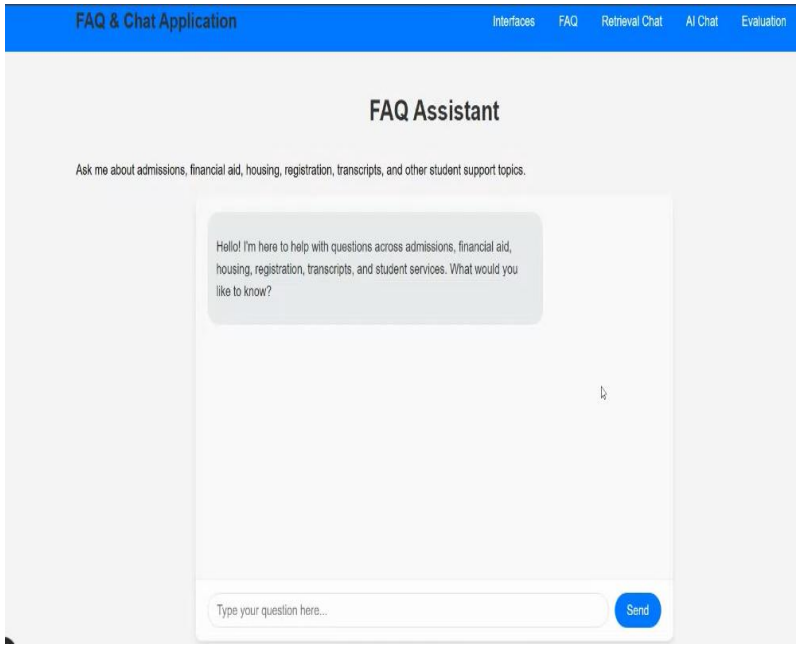
Important

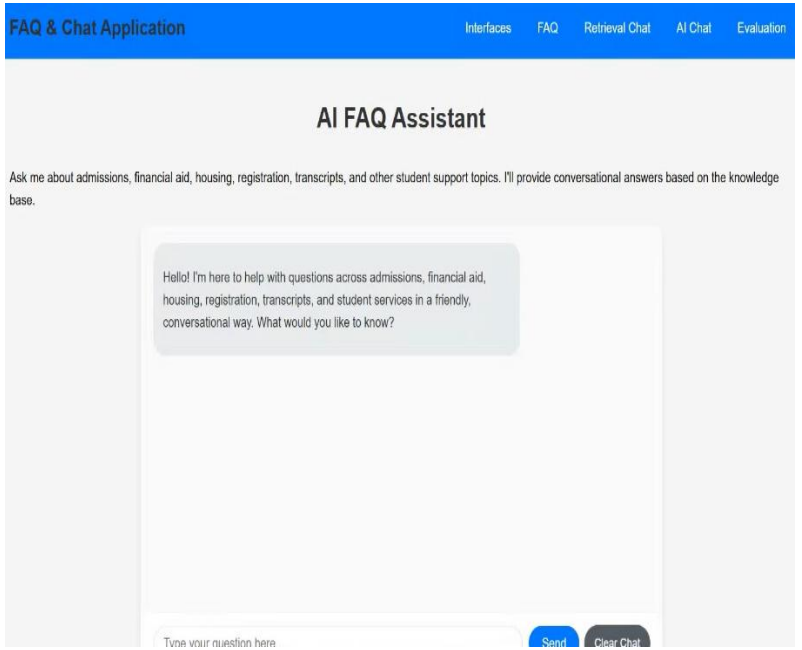
- Treat each question as a real student query
- Do not use external sources (e.g., Google)
- Try to answer as accurately and quickly as possible
- Each system contains the **same information**, presented differently

Systems Evaluation

- Accuracy and Speed

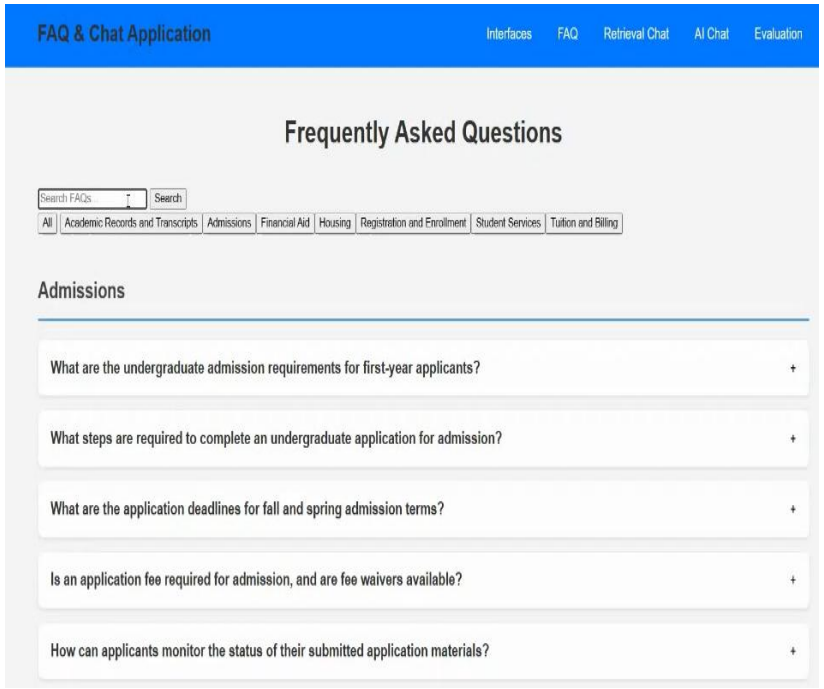


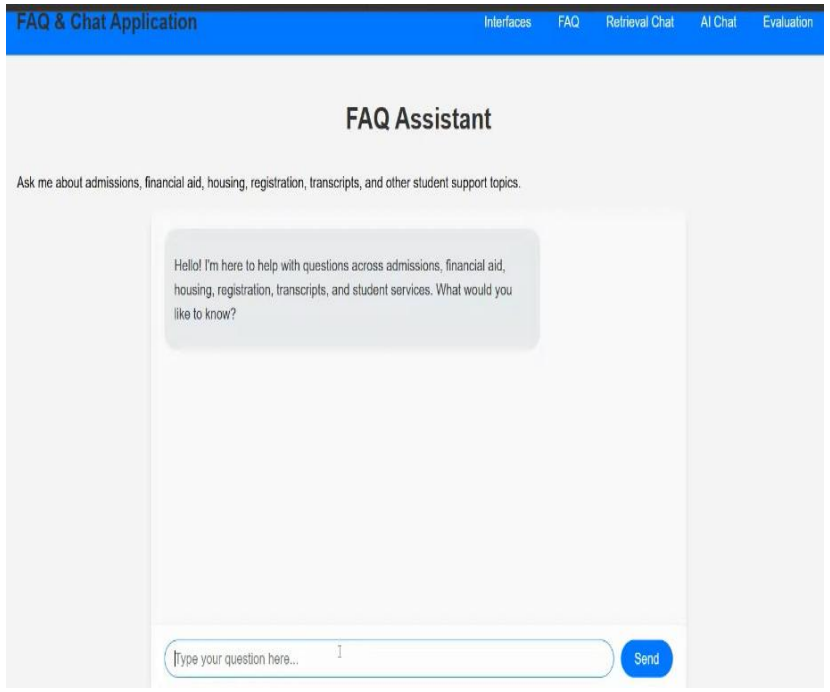


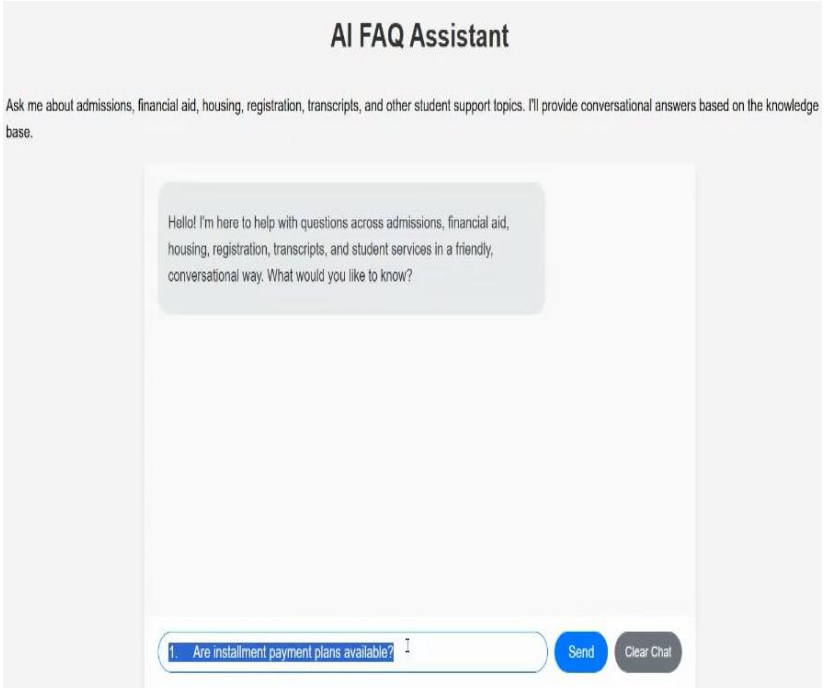


System Evaluation (Limitations)

- Intelligence and user interaction

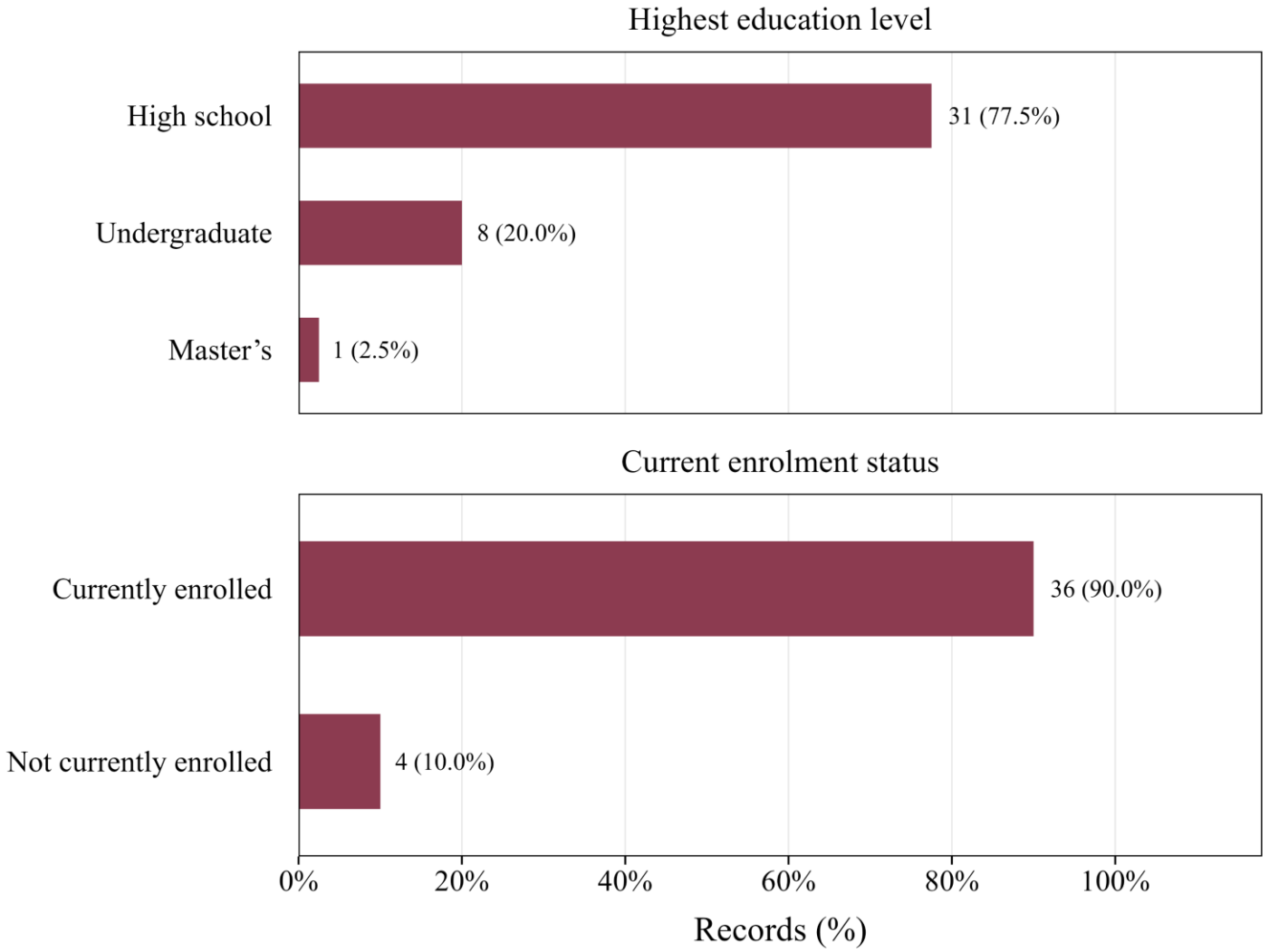






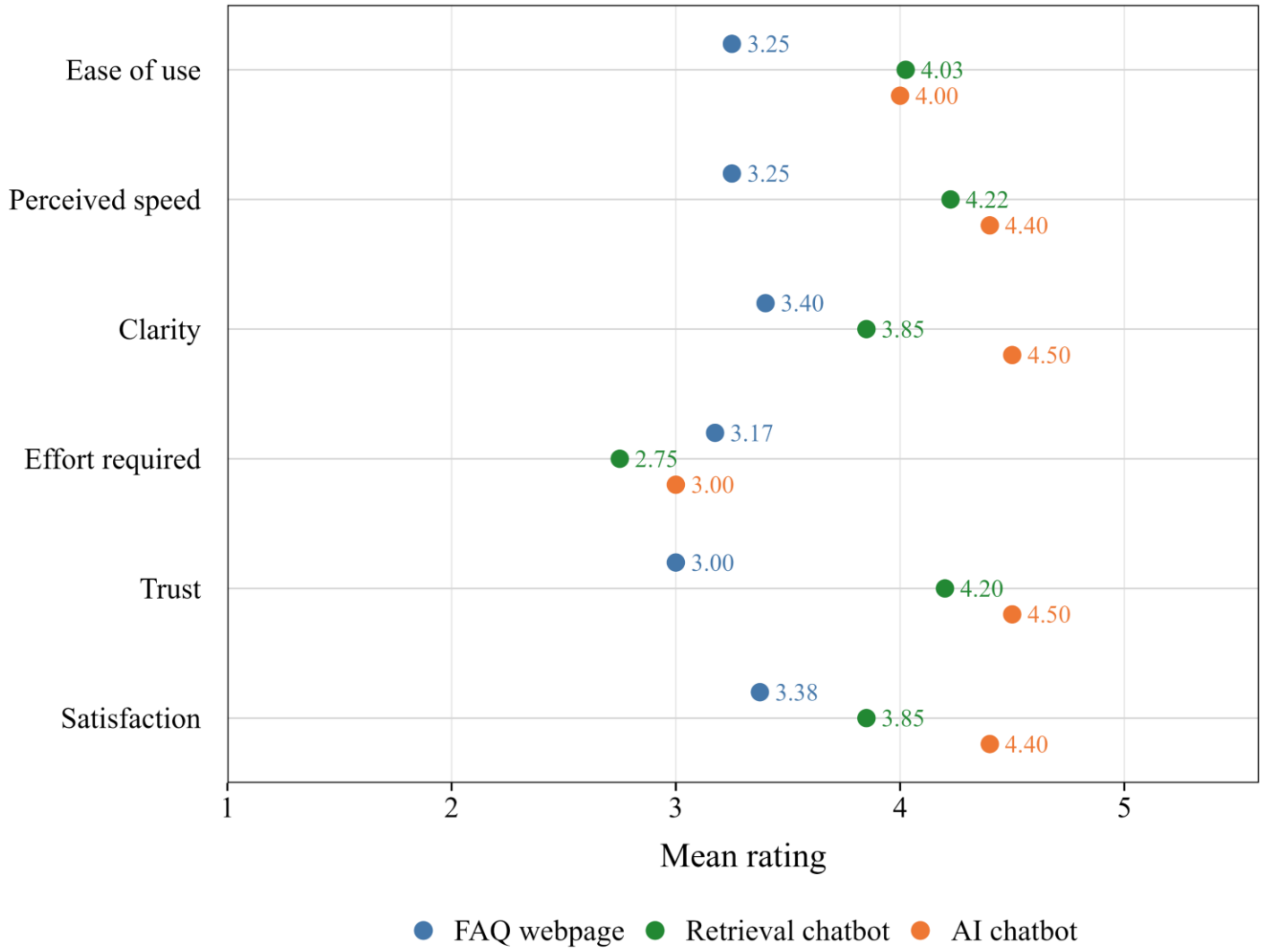
Demographics

- Of the 40 participants, 90% were currently enrolled in an educational program.
- Most had completed high school (77.5%), followed by undergraduate(20%) and master's (2.5%) degrees.



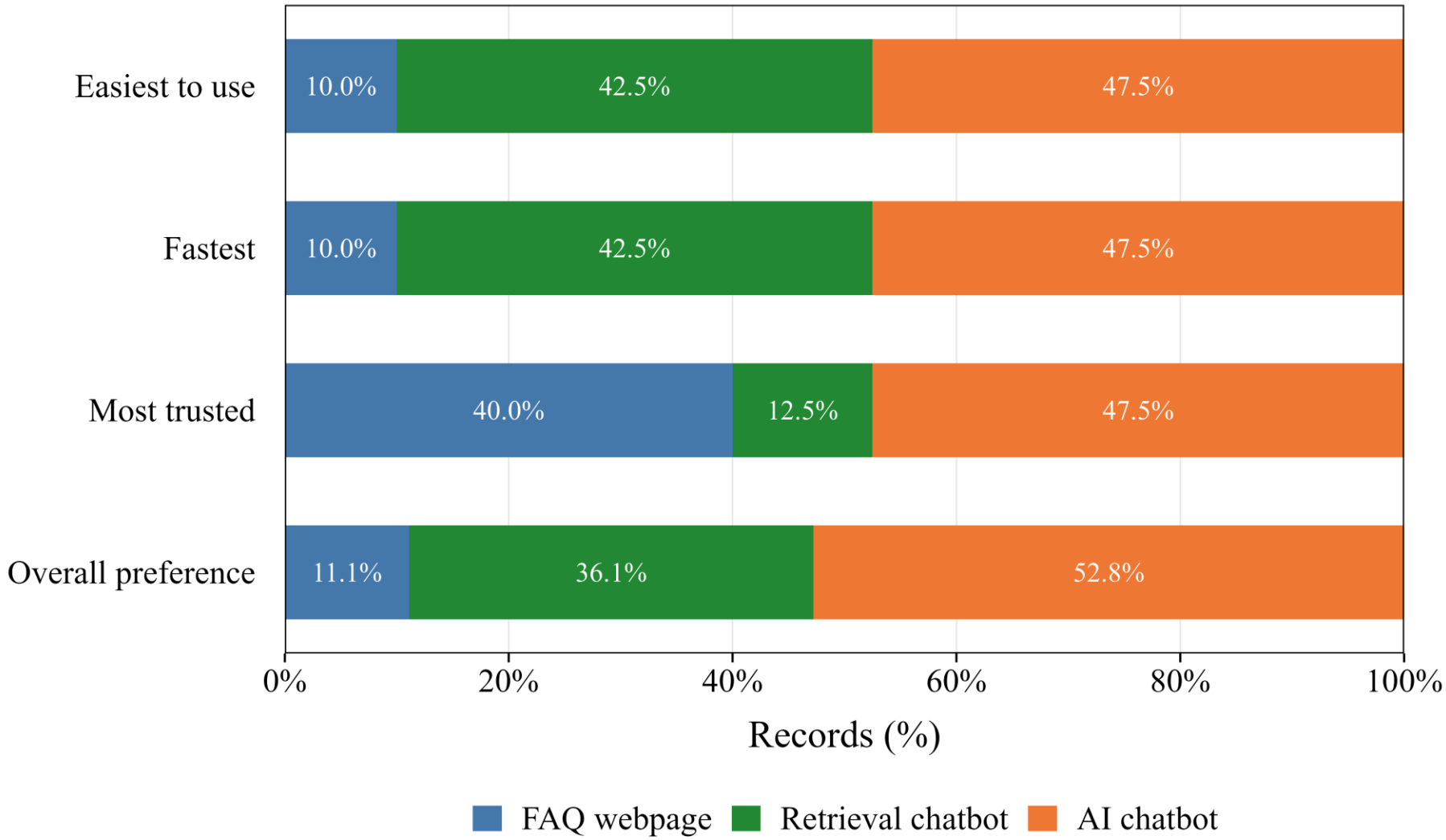
Overall User-Experience Ratings

- The retrieval chatbot was rated easiest to use and required the least reported effort, while the AI chatbot scored highest in perceived speed, clarity, trust, and satisfaction.
- Both chatbots received more favorable ratings than the FAQ webpage across most measures.



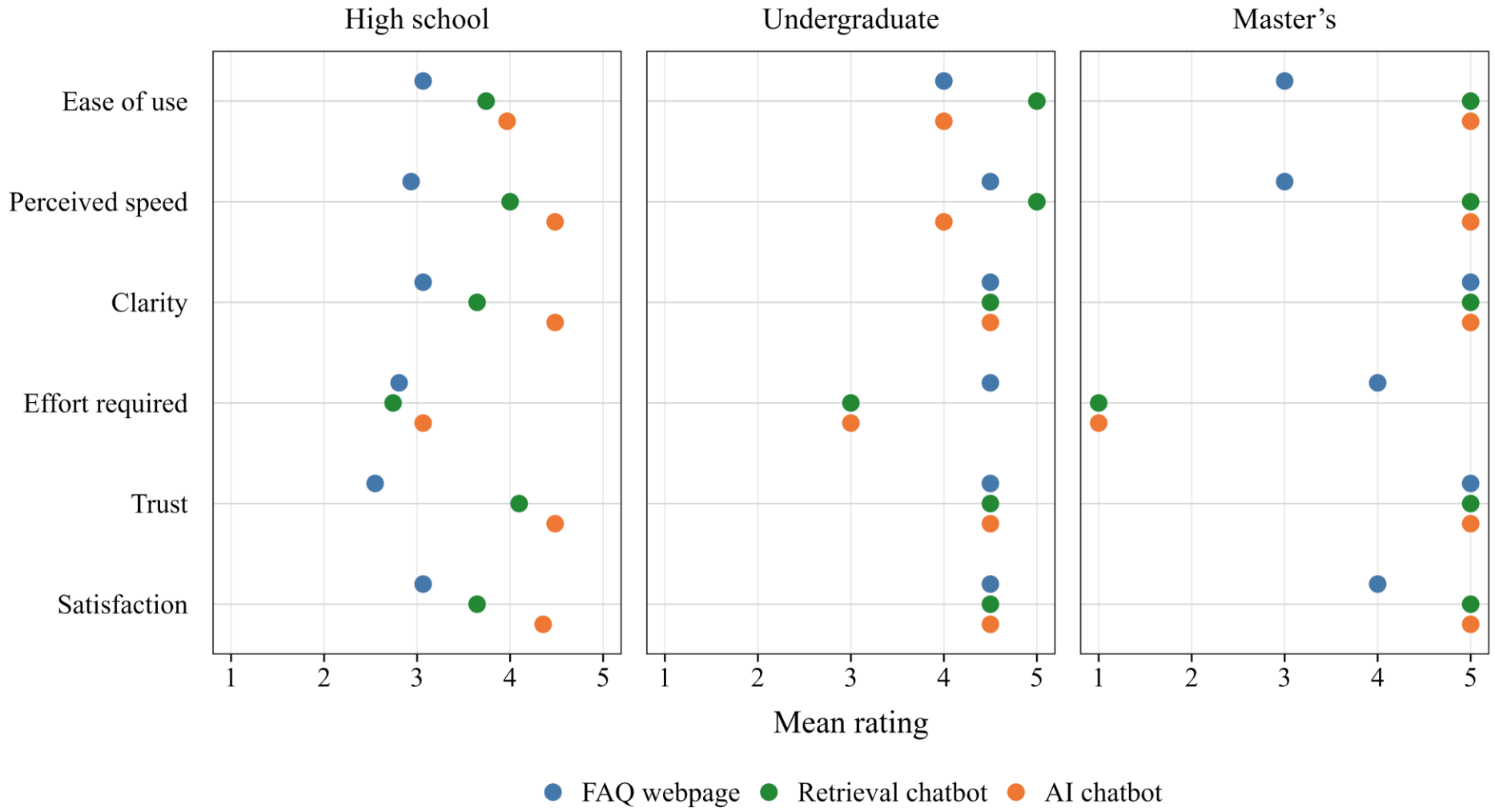
Discussion: Overall Participants' Experience

- The AI chatbot was most frequently selected as easiest to use, fastest, and most trusted (47.5% each).
- Overall, the AI chatbot was preferred by 52.8% of respondents expressing a preference, followed by retrieval(36.1%) and FAQ (11.1%).



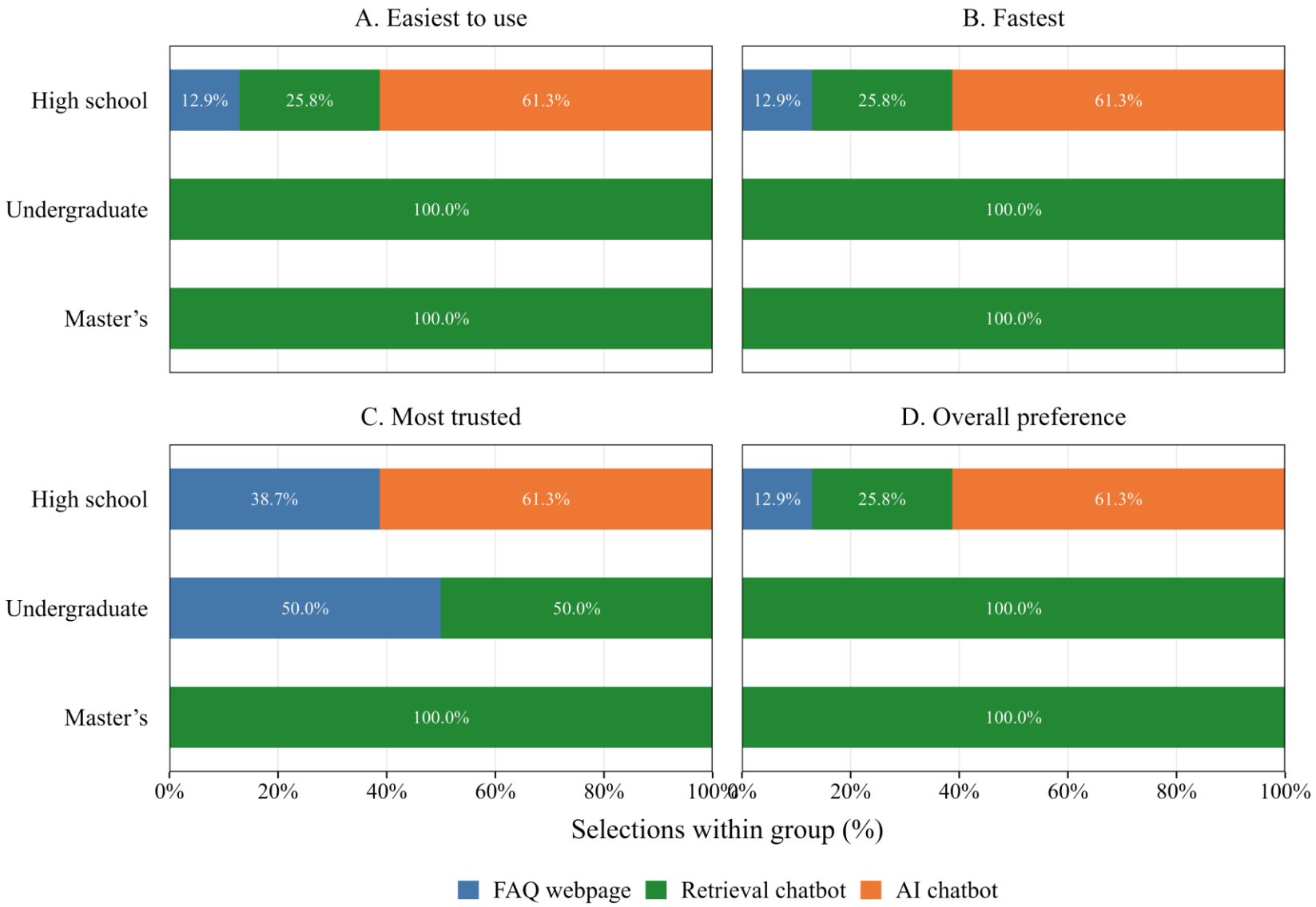
User-Experience Ratings by Education Level

- AI generally received higher ratings among high-school graduates, while retrieval showed advantages in ease of use and perceived speed among undergraduates.



Participants' Experience by Education Level

- High-school graduates generally preferred the AI chatbot, while participants with undergraduate or master's degrees favored the retrieval chatbot.



Conclusions and Future Work

Conclusions

- Both chatbots received more favorable ratings than FAQ across most measures.
- Retrieval led in ease and lower effort. AI led in perceived speed, clarity, trust, and satisfaction.
- No interface excelled across all measures.

Future work

- Use real university information and a larger, more balanced sample size for each group.
- Add validated cognitive-load and behavioral measures.

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Thank you!!!

- Scan the QR Code to participate
- Contact me at **hkddakpo@st.ug.edu.gh** for collaboration

Please Scan Me